

Next-Gen Team Builder with Predictive AI

1. Problem Description

Develop a solution that helps users create winning fantasy sports teams utilizing player data. The focus is only on cricket.

The primary task is to build a robust ML model that assists users in team creation by predicting player performance based on historical data, contextual factors (like match conditions), and other game-relevant variables. Your model should not only deliver high-performance predictions but also be accompanied by detailed explainability of features to help users understand the rationale behind player recommendations.

Beyond the model itself, your solution should include a user interface (UI). **Note on UI:** The UI holds 0% weightage in evaluation, so any basic interface will suffice.

2. Expected Output: Python-based UI with Two Interactive Interfaces

Design and develop a Python-based tool with two key interfaces:

1. **Product UI:** User-friendly team recommender.
2. **Model UI:** Model evaluation interface .

2.1 Interface 1: Product UI – Ultimate Team Selection Tool

Objective: Provide users with an interactive experience to recommend a team of 11 cricket players for a fantasy game in upcoming matches . The output should be explainable .

Inputs:

1. **Team Names:** Provide exact team names as listed in Cricsheet data .
2. **Match Date:** Provide the date of the upcoming match .
3. **Training Data Restriction:** You are to use Domestic → Men's → IPL → 2022–2025 ball-by-ball dataset. **DO NOT train on 2026 data under any circumstances**, as that data will be used strictly for testing. Non-adherence will lead to disqualification.

Important Constraints:

- **Selection Composition:** Each team must consist of 11 players from the squads of both teams :
 - Batsman: 1–8
 - Bowler: 1–8
 - All-Rounders: 1–8
 - Wicket-Keeper: 1–8

- At least 1 player from each team must be included .

Goal: Recommend the best performing team (Dream Team) that scores the highest total fantasy points based on the specified fantasy point system .

Output: Display a list of the best 11 players predicted to perform best . Each player's contribution should be justified .

2.2 Interface 2: Model UI

Objective: Allow evaluators to assess model performance across specific training and testing periods, validate predictions, view metrics, and save updated model versions .

Output:

- Train on specified training data only .
- Generate predictions for the matches in specified test duration .
- Output results in a CSV file containing: Match Date, Team 1, Team 2, Predicted Best 11 Players, Dream Team 11 Players, Predicted Points, Actual Points, and MAE (Mean Absolute Error) .
- Save retrained models in the `src/model_artifacts` folder and processed data in `src/data/processed` .

3. Judging Criteria

- **ML Model Quality & Explainability (70%):** Accuracy and robustness in predicting player performance, model explainability, feature importance, and creative usage of performance/contextual data .
- **Presentation & Documentation (30%):** Clarity in articulating the solution, structured presentation, and completeness of documentation .
- **User Interface (0%):** UI carries no weightage; a basic UI is sufficient.

4. Submission Guidelines & Rules

1. Submit code and documentation on GitHub following standard repository structures . Include a `requirements.txt` file .
2. Data sources are restricted to Cricsheet data (Domestic → Men's → IPL → 2022–2025). Do not train on 2026 data. External publicly available data (e.g., weather, pitch reports) is allowed if integrated programmatically .
3. Everything must be done programmatically .
4. Ensure feature availability for future match predictions .
5. Model predictions generated in the Model UI must be reproducible .
6. Product UI must generate team recommendations within 10 seconds max .
7. Submit a video demo and a report (up to 10 pages) detailing model architecture, features, and results .
8. Match Predictions Assumptions:

- Toss has not happened yet (no information on playing 11 or batting order) .
- Complete squad information (~ 15 members) is available .
- Venue and pitch type information are available .